

IoT-Based Health Monitoring System – Component List and Description

Must-Have Components

- **ESP32 Development Board**

Acts as the brain of the system. It collects sensor data (HR, BP, SpO₂), processes it, connects to Wi-Fi, and sends data to the server or alerts via email/WhatsApp.

- **MAX30102 Pulse Oximeter Sensor**

Measures heart rate (HR) and blood oxygen saturation (SpO₂). Key for detecting conditions like tachycardia, bradycardia, or low oxygen levels.

- **Blood Pressure Sensor Module**

Monitors systolic and diastolic blood pressure. Helps in detecting hypertension or hypotension. Can be replaced with a digital cuff if unavailable.

- **MPU6050 Accelerometer + Gyroscope Module**

Detects patient movement and sudden falls. Useful for fall detection and motion tracking.

- **Active Buzzer Module**

Produces an audible alarm when abnormal vitals are detected to alert the patient immediately.

- **LEDs + 220Ω Resistors**

LEDs indicate the patient's health status (e.g., green = normal, red = abnormal). Resistors protect LEDs from excess current.

- **Rechargeable Li-ion / Li-Po Battery (3.7V)**

Provides portable power to the entire system, making it wearable and wireless.

- **TP4056 Battery Charging Module**

Allows safe charging of the Li-ion/Li-Po battery via USB. Includes overcharge protection.

- **Battery Holder (for 18650 cells)**

Holds the 18650 battery securely and connects it to the charging circuit.

- **Jumper Wires (Male–Male, Male–Female)**

Used to connect sensors and modules to the ESP32 on the breadboard.

- **USB Cable (for ESP32)**

Used to program and power the ESP32 through a computer or USB adapter.

Recommended Components

- **• 0.96" or 1.3" OLED Display (I2C, SSD1306)**

Displays live readings of HR, BP, and SpO₂ values directly on the device.

- **• Push Buttons**

Used for system reset or manual emergency alert triggering.

- **• GPS Module (NEO-6M or NEO-7M)**

Provides accurate live location coordinates to include in emergency alerts.

- **• SIM800L / SIM900 GSM Module**

Enables SMS/call alerts even without internet or Wi-Fi access.

- **• Velcro Strap / Wristband**

Used to make the system wearable for the patient.

- **• Small Enclosure / Case**

Houses all electronic components safely in a compact form.

- **• Multimeter**

Used for debugging and verifying voltage/current readings during setup.

- **• Extra Jumper Wires and Resistors**

Backup components for testing and replacing faulty connections.

- **• Breadboard**

Used for circuit prototyping without soldering. Allows flexible wiring and quick changes during testing.

IoT Health Monitoring System – Components List (with Quantity)

Core Controller

Component	Qty	Purpose
ESP32 Development Board (with Wi-Fi + Bluetooth)	1	Main microcontroller for sensor data collection and communication with Flask app/web backend.

Sensors

Component	Qty	Purpose
MAX30102 Pulse Oximeter Sensor	1	Measures Heart Rate (HR) and SpO ₂ .
MAX30102 Pulse Oximeter Sensor	1	Measures systolic/diastolic blood pressure (optional but adds accuracy).
MPU6050 Accelerometer + Gyroscope Module	1	Detects motion and falls.

Alerts & Indicators

Component	Qty	Purpose
Active Buzzer Module	1	Gives audible alarm during emergencies.
LED (Red, Green)	2 (1 each)	Indicates status (Normal/Abnormal).
220Ω Resistors	2	Used with LEDs to prevent overcurrent.

Display & User Interface

Component	Qty	Purpose
0.96" or 1.3" OLED Display (SSD1306, I2C)	1	Displays live HR, BP, and SpO ₂ readings.
Push Button (for reset/emergency trigger)	2	Manual emergency alert or reset function.

Connectivity

Component	Qty	Purpose
NEO-6M GPS Module	1	Provides live location data for alerts.
SIM800L / SIM900 GSM Module	1	Enables SMS/call alerts without Wi-Fi.

Power Supply

Component	Qty	Purpose
Rechargeable Li-ion Battery (3.7V, 18650 or Li-Po)	1	Powers the wearable device.
TP4056 Battery Charging Module	1	Safely charges Li-ion battery via USB.
Battery Holder (for 18650)	1	Holds the Li-ion battery securely.

Prototyping Materials

Component	Qty	Purpose
Breadboard	1	For initial circuit testing (you already have this).
Jumper Wires (Male–Male, Male–Female)	~30	Connections between sensors and ESP32.
USB Cable (Type-C or Micro USB, depending on ESP32)	1	For programming and powering the ESP32.

Wearable Assembly

Component	Qty	Purpose
Velcro Strap / Wristband	1	Makes the device wearable.
Small Plastic or 3D Printed Enclosure	1	To house ESP32 and sensors neatly.

Estimated Prices

Component	Approx Price*
ESP32 development board	₹280–₹600
MAX30102 Pulse Oximeter Sensor Module	₹100–₹500
Breadboard + jumper wires	(you said you already have breadboard)
Battery & Charging Module (Li-ion + TP4056)	~₹150–₹300 for module + battery
Buzzer / LEDs / Resistors	~₹20–₹100 depending on quality and quantity